Modeling and Predicting Chicago Traffic Volume with SARIMA and ARIMAX

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1 Abstract

Accurate forecasting of traffic volume is essential in large metro areas for its effect on urban planning, traffic control measures, infrastructure, and GPS routing applications. This project analyzes daily traffic counts in the Chicago Metro area from 2015-2017 to model and predict short-term and long-term traffic patterns. The primary objective is to evaluate the effectiveness of using times series models, specifically Seasonal Autoregressive Integrated Moving Average (SARIMA) and Autoregressive Integrated Moving Average with exogenous variables (ARIMAX), in capturing series dependencies and improving forecast accuracy. Practical uses for this project extend from short-term applications in rerouting GPS guidance applications to long-term applications in the realms of infrastructure, traffic signal timing, and law enforcement presence.

2 Introduction

2.1 Background

One of the few certainties of urban living is dealing with traffic congestion and delays. Since the 1980's, the average time spent in traffic congestion has over tripled and continues to increase at a rate of 17% every 5 years. This causes headache, wasted fuel, and increased urban pollution along with stress on infrastructure, public transportation, and emergency services [1]. Urban population continues to grow, with urban population expected to rise from 55% to 68% by 2050. This equates to a stress of over 2 billion additional people in urban areas [2].

With this increase in urban population, traffic congestion will only be expected to grow. To better plan and account for this, one may want to develop a method of predicting traffic influxes along with understanding the structure and pattern that drives it. We will seek to accomplish this using traditional methods of time series, exploratory, and regression analysis. We will examine seasonality and stationarity structure, autoregressive and moving average dependence, and exogenous variables to decode traffic trends.

2.2 Research Objectives

Our primary objective for the study is to determine what drives traffic levels in the goal of developing an accurate prediction model. Natural extensions of this objective is to:

1. Identify seasonality trends
2. Quantify exogenous variable effects
3. Develop an accurate prediction model

3 Materials and Methods

3.1 Study Context

Data was collected from reports done by the City of Chicago from October 28, 2015 to February 12, 2017. The study focused on downtown areas and provides daily count values. The study can be defined as observational rather than experimental as observations were recorded via pneumatic sensor tubes with no external influence or containment of natural traffic flow. We will hold out the final 30 days of the dataset for a cross-validation and retain the remaining data for model fitting. This will allow us to examine the generalization of our model to future unseen data.

3.2 Explanatory Variables

Our explanatory variables are divided into three primary groups: time related, event related, and climate related. Time related variables are the primary object of our analysis. Most of the information gained from these variables will be in seasonality and trend analysis.

Event related variables will likely be one of the more important variables in determining extreme traffic fluctuations, likely due to traffic associated with major sporting, entertainment, and holiday events. Climate related variables may help us to fine tune more minor fluctuations in traffic counts related to changes in driving conditions. These exogenous variables will be considered only in the ARIMAX extension of our time series analysis where we will combine traditional ARIMA/SARIMA models with linear regression approaches.

3.3 Response Variable

Naturally, our response variable will be traffic counts as determined by the method of using pneumatic sensors placed on the road. This variable will be measured in total vehicles traveled across the main causeway.

4 Statistical Methods

In this section, we will present the primary modeling methods that we will utilize to capture the structure of our data. Auto regressive and moving average methods and their extensions dominate traditional time series analysis. We will examine each of these and how they may be applied to our data.

4.1 Auto Regressive Model

The idea of auto regressive modeling was first introduced by Yule (1927). The auto regressive model of order p, denoted as AR(p), explains an observation x_t as a linear combination of past observations. We can formally define this model as:

$$x_t = \phi_1 x_{t-1} + ... + \phi_p x_{t-p} + w_t = \sum_{i=1}^{p} \phi_i x_{t-i} + w_t \quad (1)$$

where each ϕ_i is an auto regressive coefficient and w_t is a white noise term such that $w_t \overset{\text{iid}}{\sim} \mathcal{WN}(0, \sigma_w^2)$. For higher order AR(p) processes, we can utilize the AR polynomial to interpret the nature of the process. The AR polynomial can be written as:

$$\phi(z) = 1 - \phi_1 z - ... - \phi_p z^p \quad (2)$$

and we can solve for the polynomial roots by setting it equal to 0. The AR process is stationary if all roots lie outside the unit circle, i.e. $|z_i| > 1$ for every root z_i. Roots represent all the coefficients jointly so, intuitively, their inverse can be viewed similarly to the coefficient of the AR(1) process. In this manner, roots with magnitude greater than 1 are analogous to $|\phi_1| < 1$ in the AR(1) process and will therefore force the effect of past observations to die out.

4.2 Moving Average Model

The moving average model is the other primary base model utilized in time series analysis and draws its origins from Slutzky (1937). A moving average model of order q is often denoted as MA(q), and explains the current observation as a linear combination of the current and past shocks such that:

$$x_t = w_t + \theta_1 + ... + \theta_q w_{t-q} = \sum_{j=0}^{q} \theta_j w_{t-j} \quad (3)$$

where each θ_i is the coefficients for the given shock and w_t is once again a standard Gaussian white noise term. The moving average process is fundamentally different than the AR(q) process because it recalls only a finite set of past effects while the AR process recursively recalls observations infinitely. Like the AR process, the MA process can be written as:

$$\theta(z) = 1 + \theta_1 z + ... + \theta_q z^q \quad (4)$$

While the roots of the AR polynomial signal stationarity, the MA polynomial determines invertibility. If $|z_i| > 1$ for each MA root z_i, we can recover the unobserved past shocks uniquely based on the same logic we used for the AR polynomials.

4.3 ARIMA Models

A natural extension of the two previous models discussed is their combination into one singular model, referred to as an ARMA or ARIMA (for non-stationary series) written as $\phi(z)x_t = \theta(z)w_t$. An ARIMA model is often denoted as:

$$ARIMA(p, d, q) \quad (5)$$

where p is the order of the auto regressive portion, d is the degree of differencing, and q is the order of the moving average portion. An ARIMA model can be extended to include seasonality, turning the model into what is known as a seasonal ARIMA (SARIMA) model. A SARIMA model is denoted as:

$$SARIMA(p, d, q), (P, D, Q)[s] \quad (6)$$

where P, D, Q are the seasonal forms of p, d, q and s is the period of seasonality. Because of the weekly structure of our data, it is likely that we will end up using a SARIMA model with $s = 7$. Formally, we represent the equation for the SARIMA model as:

$$\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D X_t = \Theta(B^s)\theta(B)W_t \quad (7)$$

4.4 ARIMAX

One final extension of our models is the auto regressive integrated moving average with exogenous inputs (ARIMAX) model. Since our dataset provides additional non-time related variables such as weather and event information, we can incorporate this into our model via the use of linear regression. Defined as:

$$x_t = \beta_0 + \beta' Z_t + N_t \quad (8)$$

where β is the vector of regression coefficients, Z_t is the model matrix, and N_t is the ARIMA or SARIMA portion. This gives us:

$$\phi(B)(1-B)^d N_t = \theta(B)w_t \quad (9)$$

This will likely be the best model for our purposes due to the seasonal nature of our data and the exogenous predictor variables provided, but we will explore this further in our exploratory analysis and model selection.

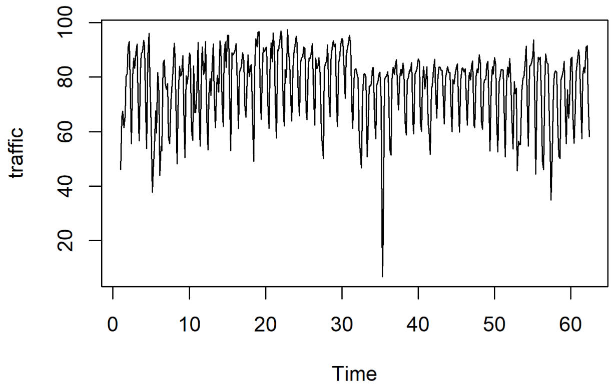


Figure 1: Raw traffic counts for the Chicago metro.

5 Exploratory Data Analysis

We will begin by analyzing the time series as a whole and see if we can uncover any initial information. As can be seen in the raw traffic counts plot in Figure 1, there is no immediately decipherable overall trend in traffic volume. We do notice, however, that the time series seems to be dominated by seasonality with a weekly periodicity. One may also note that there appears to be a larger variance in traffic during the early and late weeks, corresponding to winter months in which driving conditions fluctuate more significantly.

Examining the ACF and PACF of the raw traffic counts, we can note a few things. First the ACF demonstrates a strong weekly seasonal pattern with lags 7, 14, 21,... showing significant spikes. We may then conclude that seasonal differencing is necessary. The PACF also shows significant lag dependence going out approximately one week. Low-order auto regressive terms may help to decipher this dependence. Overall, we may consider a seasonal auto regressive moving average (SARIMA) model based on the inference from the time series, ACF, and PACF plots. We will explore this further utilizing differencing and other transformations.

5.1 Log and Box-Cox Transformations

In some occasions, applying a transformation to the response can help to:

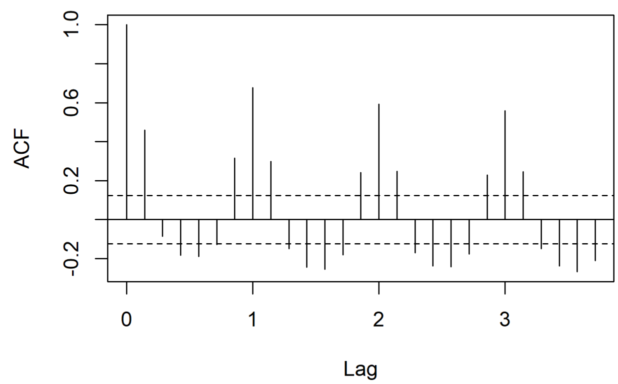
1. Stabilize the variance across the time series (we witnessed changing variances across winter and summer months).

2. Convert multiplicative effects into additive effects (keeping in line with ARIMA structure).

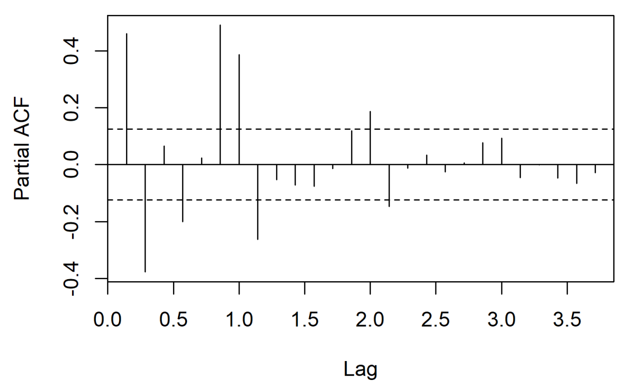
3. Help achieve stationarity along with differencing.

Transformations commonly utilized in time series data include logarithmic and Box-Cox transformations, so we will begin here. We should note that we will also perform a first order differencing to the data after taking the transformation

The logarithmic transformation is defined simply as taking $\log(x_t)$. The Box-Cox transformation, originally proposed by



(a) Raw counts ACF.



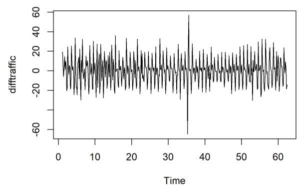
(b) Raw counts PACF.

Box (1964), allows for a more flexible transformation that depends on the nature of the data. It has a slightly more complex structure and can be defined formally as:

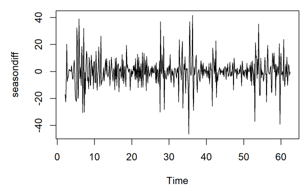
$$Y^{(\lambda)} = \begin{cases} \dfrac{Y^{\lambda} - 1}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases} \tag{10}$$

for some λ such that the log-likelihood is maximized, shown as $\hat{\lambda} = \arg\max_{\lambda} \ell(\lambda)$.

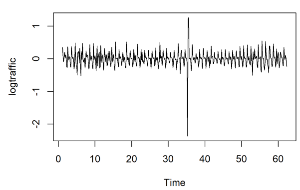
Though not shown, these transformations failed to show significant improvement when performing diagnostics on the time series, ACF, and PACF. Unfortunately, it appears that neither transformation was able to significantly reduce the seasonality structure or the short lag dependence. We would be wrong to assume that simply applying these transformations converge our time series towards stationarity, and will therefore proceed with utilizing other methods.



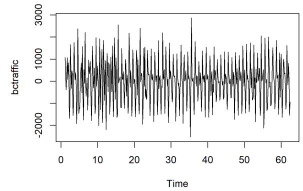
(a) Differenced counts.



(b) Seasonally differenced counts

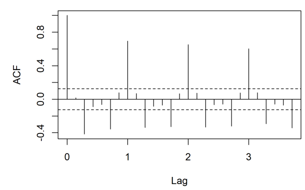


(c) Log transformed counts.

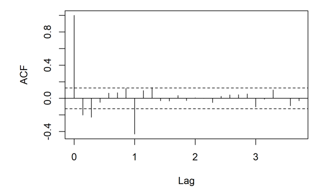


(d) Box-Cox transformed counts

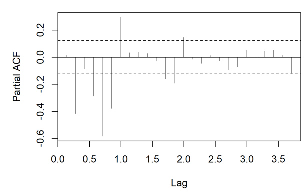
Figure 3: Time series of original, differenced, log transformed, and Box-Cox transformed traffic counts.



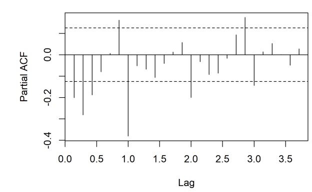
(a) Differenced ACF



(b) Seasonly Differenced ACF



(c) Differenced PACF



(d) Seasonly Differenced PACF

Figure 4: ACF and PACF of differenced (left) and seasonly differenced traffic counts (right)

5.2 Differencing

We will first begin by performing a standard differencing on the data, effectively transforming the data from raw observations to the difference of observations. We denote this transformation simply as:

$$\nabla x_t = x_t - x_{t-1} \tag{11}$$

Differencing the data helps to stabilize the mean of our time series by removing trends. We can further examine the effect that differencing has in more detail by considering the ACF and PACF of the new time series. The left side of Figure 4. shows the resulting plots. The ACF demonstrates a still significant dependence on past observations, especially at weekly lags.

This may suggest to us the importance of utilizing a higher order differencing equation, known as seasonal differencing. Naturally, we will choose the order of the differencing to be 7 to reflect the natural weekly cycle of the data. It is standard to perform the original single order differencing to the data after the seasonal differencing. We define this form of differencing transformation as:

$$\nabla\nabla^7 x_t = (1-B)(1-B^7)x_t = (x_t - x_{t-1})(x_t - x_{t-7}) \tag{12}$$

The right side of Figure 3. shows the ACF and PACF of the seasonaly differenced data. We witness a significant reduction in autocorrelation between current and past observations. There is still a significant dependence at the weekly lag, but only out to a single week.

Both PACF plots show little effect from the regular or seasonal differencing.

6 Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller Test, introduced by Dickey and Fuller (1981) to expand on their original test from Dickey and Fuller (1979), tests to see if the series contains a unit root, indicating a nonstationary time series. In other words, the ADF test equates to a test for stationarity. The test is based on the regression:

$$\nabla x_t = \alpha + \beta t + \gamma x_{t-1} + \sum_{i=1}^{p} \delta_i \nabla x_{t-i} + \epsilon_t \tag{13}$$

We perform this test on the original time series in addition to the transformations we used in the previous sections.

Table 1: Augmented Dickey–Fuller Test Results for Traffic Series Transformations

Series	Dickey–Fuller Statistic	Lag	p-value
Traffic	-4.3925	7	< 0.01
∇log(Traffic)	-18.7870	7	< 0.01
∇ Box-Cox	-19.8140	7	< 0.01
∇x_t	-10.9960	7	< 0.01
$\nabla\nabla^7 x_t$	-11.2070	7	< 0.01

Note: The null hypothesis of the Augmented Dickey–Fuller (ADF) test is that the series contains a unit root (nonstationary). All reported p-values were smaller than the printed threshold of 0.01, indicating strong evidence against the null hypothesis of nonstationarity.

The ADF statistics for each time series suggests stationarity.

7 Model Selection

We can now move forward with considering various different models. Our exploratory analysis suggests that our series contains both AR and MA dependence, so we will naturally move forward with fitting a SARIMA model.

7.1 SARIMA Model

Using the auto.arima function in r, we obtain the best model parameters for our model such that:

Component	Parameter	Estimate	Standard Error
NS MA	$\hat{\theta}_1$	0.5992	0.0497
NS MA	$\hat{\theta}_2$	0.2396	0.0589
NS MA	$\hat{\theta}_3$	0.1052	0.0552
NS MA	$\hat{\theta}_4$	0.1037	0.0465
S MA	$\hat{\Theta}_1$	-0.8993	0.0442
–	Var (σ^2)	55.45	–

Table 2: Estimated Parameters for the Fitted SARIMA$(0,0,4)(0,1,1)_7$ Model

with the following nonseasonal and seasonal MA polynomials:

$$\theta(z) = 1 + 0.599z + 0.240z^2 + 0.105z^3 + 0.104z^4$$

$$\Theta(z^7) = 1 - 0.899z^7$$

which gives nonseasonal MA roots

$$z_1, z_2 = -1.286 \pm 0.942i$$

$$z_3, z_4 = 0.781 \pm 1.781i$$

and seven seasonal MA roots that share a common modulus

$$|z| = 1.015$$

The seasonal and all four non-seasonal roots fall outside of the unit circle and therefore both processes are invertible. This means that the current and past observations contain enough stable information to reconstruct the underlying random shocks. In the specific case of our traffic data, we interpret shocks to be the impact of unexpected automobile accidents, weather, congestion spikes, etc. and we therefore can go back to uncover this information from the current observation. However, the roots of the seasonal MA polynomial are very close to falling on the unit circle, and therefore will be unreliable for recovering past information.

From the model parameters, we can infer that positive disturbances on traffic rates from the preceding days will contribute further positive disturbances on the current day and vice versa. These disturbance effects quickly die out after the day immediately before the current observation. From the seasonal MA coefficients, we note that positive disturbances from the previous week have the opposite effects, contributing negative disturbances to the current observations and vice versa.

One could intuitively decipher both the seasonal and non-seasonal coefficients by thinking about the impact of a severe weather phenomenon, say a large blizzard. As the blizzard arrives, it makes travel very difficult for the day or two during which the blizzard lasts. This effect will die off over the next few days as the roads slowly improve. However, due to the significant reduction in traffic as people put off running errands and traveling, we can expect these same people to complete their missed tasks the following week, thus increasing traffic volume.

7.2 ARIMAX Model

The final extension of time series model is the ARIMAX model which extends the purely time related SARIMA model to incorporate exogenous predictor variables. We will utilize the same SARIMA order as before but now include the regression adjustment, obtaining the following model parameters:

Component	Parameter	Estimate	Standard Error
NS MA	$\hat{\theta}_1$	0.6268	0.0503
NS MA	$\hat{\theta}_2$	0.2259	0.0591
NS MA	$\hat{\theta}_3$	0.1081	0.0535
NS MA	$\hat{\theta}_4$	0.1092	0.0460
S MA	$\hat{\Theta}_1$	-0.8833	0.0442
Regression	$\hat{\beta}_{holiday}$	-17.3155	1.5044
Regression	$\hat{\beta}_{snow}$	-51.2337	41.8219
–	Var (σ^2)	42.02	–

Table 3: Estimated Parameters for the Fitted ARIMAX Model with SARIMA$(0,0,4)(0,1,1)_7$ Errors

with the following nonseasonal and seasonal MA polynomials:

$$\theta(z) = 1 + 0.627z + 0.226z^2 + 0.108z^3 + 0.109z^4$$

$$\Theta(z^7) = 1 - 0.883z$$

which gives nonseasonal MA roots

$$z_1, z_2 = -1.292 \pm 0.885i$$

$$z_3, z_4 = 0.797 \pm 1.761i$$

and seven seasonal MA roots that share a common modulus

$$|z| = 1.018$$

Like in the case of the SARIMA model, all of the MA polynomial roots fall outside of the unit circle, and we can therefore recover information about the past shocks from our current observation. Once again, however, our seasonal MA polynomial roots are very close to lying on the unit circle making inference more difficult. The interpretation of these coeefficients is essentially equivelant as in the pure SARIMA model.

We will quickly decipher the additional regression coefficients. We can see that both the presence of a holiday or snow significantly reduces the traffic count, with the presence of a holiday reducing the counts much more consistently.

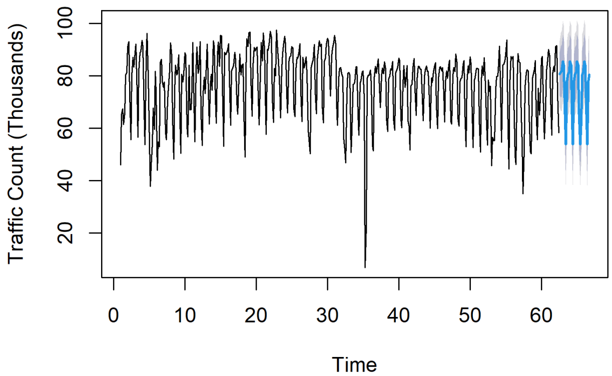


Figure 5: Predicted traffic counts with 80% and 95% confidence bounds.

7.3 Model Comparison

We recall that before fitting the models, we held out the final 30 days of the data as a test set in order to analyze the performance of each model on unseen data. We will use the accuracy on this data as the primary metric for our model comparison. We will also consider the AIC of each model in order to assess the balance between accuracy and complexity.

Model	Test/Train	RMSE	MAE	AIC
SARIMA	Test	7.528	6.571	–
ARIMAX	Test	7.231	6.410	–
SARIMA	Train	7.342	4.729	2,924.794
ARIMAX	Train	6.376	4.380	2,808.284

Table 4: Model comparison statistics (SARIMA vs. ARIMAX)

We can see in Table 4. that the ARIMAX model consistently outperforms the SARIMA model in terms of accuracy on both the training and test sets, producing lower RMSE and MAE values. In addition, the added complexity of the ARIMAX model attributed to the incorporation of a regression component does not seem to outweigh the predictive accuracy of the model as the AIC value is significantly lower for the ARIMAX model. We noted before that the inference gained from each model in terms of the AR and MA processes was very similar so we will select the ARIMAX model for its increase in predictive accuracy.

8 Prediction

Finally, we can utilize our model as a tool for traffic count prediction. Our model is able to continually predict traffic counts given we have data traffic count data from the previous week and forecast data for the date we wish to predict

Figure 5. demonstrates how the model can be used to predict future traffic counts. From visually analyzing the prediction plot, it appears that the model does a good job of capturing the seasonal trend, but possibly generalizes the traffic counts too much. This may indicate to us that there are other very important variables that are not being included into the model, or that the traffic counts are very dependent on the random nature of white noise.

9 Limitations

While our model does provide us with a much better method of predicting the traffic counts than simply relying on the average traffic volume, largely due to the seasonal structure we detected, there are still some major limitations. The primary limitation is in the inability of our model to predict the extreme fluctuations in traffic. The ability of our model to do this could be improved by the incorporation of more extreme event indicators, such as the presence of sporting/entertainment events, refining observations to specific commercial/residential/industrial districts, etc. It is also possible that the traffic volume includes a vary large random component. When we fit our SARIMA and ARIMAX models, there was a relatively high variance associated with the observations, indicating a high dependence on randomness or unobserved variables.

The secondary limitation in our model is the length of the study. Since the data was collected across a time period of just over one year, we are not able to accurately decipher the higher order monthly or yearly seasonality. There is likely a significant difference in the traffic patterns across different months and season, so in future studies, we would like to expand our observations to capture this.

10 Conclusion

Overall, we were able to detect a strong influence of weekly periodicity in our data. Utilizing seasonal differencing and MA processes, we were able to determine that traffic volume was very dependent on the day of week and the impact of previous disturbances, with concurrent effects from immediate disturbances and reciprocal effects from disturbances in the last week. We were also able to generalize our predictions to unseen traffic data, maintaining the caveat that our model was unlikely to predict extreme fluctuations in traffic counts.

References

[1] Schrank, D., Albert, L., Jha, K., and Eisele, B. (2025). *2025 Urban Mobility Report*. Texas A&M Transportation Institute, College Station, TX. Published with cooperation from INRIX. Sponsored by the Texas Department of Transportation.

[2] United Nations Department of Economic and Social Affairs, Population Division. *World Urbanization*

Prospects: The 2018 Revision. United Nations, New York, 2018.

[3] Yule, G. U. (1927). "On a Method of Investigating Periodicities in Disturbed Series, with Special Reference to Wolfer's Sunspot Numbers." *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, 226(636–646), 267–298. doi:10.1098/rsta.1927.0007

[4] Slutzky, E. (1937). "The Summation of Random Causes as the Source of Cyclic Processes." *Econometrica*, 5(2), 105–146.

[5] Box, G. E. P., and Cox, D. R. (1964). "An Analysis of Transformations." *Journal of the Royal Statistical Society: Series B (Methodological)*, 26(2), 211–243. doi:10.1111/j.2517-6161.1964.tb00553.x